

Weekly Topical Worksheet Pack**CANDIDATE****Urwah****SUBJECT****Chemistry -- Cambridge International A Level (9701)****TOTAL MARKS (WEEK)****25 marks****HOW TO USE THIS PACK**

One short set of questions per session, matching that day's topic from the schedule.

Answer in the spaces / on the lines provided. Marks for each part are shown in [brackets].

Mark schemes for the whole week are collected at the end of the pack.

These are original questions written to match Cambridge's exact style and command words — not reproductions of real past paper text.

THIS WEEK AT A GLANCE

TOPIC 22 — TOPIC 22: ANALYTICAL TECHNIQUES (25 marks)

MENTORA ACADEMY · EXCELLENCE IN EDUCATION.**TOPICAL PRACTICE — WEEK 4****Analysis -- Topic 22: Analytical Techniques (IR & MS)**

AS Level Chemistry 9701 -- 2025-2027 Syllabus

Question Pack

TOPIC 22 — TOPIC 22: ANALYTICAL TECHNIQUES

1 Analytical Techniques — 9701/21/O/N/25/Q1(e) Syllabus 22.1

Question 1 on Analytical Techniques. Fig. 2.1 shows an IR spectrum.

- (a) Identify the functional group responsible for the broad absorption at 3200–3600 cm⁻¹.

[1]

2 Analytical Techniques — 9701/21/O/N/25/Q2(e) Syllabus 22.1

Question 2 on Analytical Techniques. Fig. 2.1 shows an IR spectrum.

- (a) State the significance of the molecular ion peak in a mass spectrum.

[1]

3 Analytical Techniques — 9701/21/O/N/25/Q3(e) Syllabus 22.1

Question 3 on Analytical Techniques. Fig. 2.1 shows an IR spectrum.

- (a) Identify the functional group responsible for the broad absorption at 3200–3600 cm⁻¹.

[1]

4 Analytical Techniques — 9701/21/O/N/25/Q4(e) Syllabus 22.1

Question 4 on Analytical Techniques. Fig. 2.1 shows an IR spectrum.

- (a) State the significance of the molecular ion peak in a mass spectrum.

[1]

5 Analytical Techniques — 9701/21/O/N/25/Q5(e) Syllabus 22.1

Question 5 on Analytical Techniques. Fig. 2.1 shows an IR spectrum.

- (a) Identify the functional group responsible for the broad absorption at 3200–3600 cm⁻¹.

[1]

6 Analytical Techniques — 9701/21/O/N/25/Q6(e) Syllabus 22.1

Question 6 on Analytical Techniques. Fig. 2.1 shows an IR spectrum.

- (a) State the significance of the molecular ion peak in a mass spectrum.

[1]

7 Analytical Techniques — 9701/21/O/N/25/Q7(e) Syllabus 22.1

Question 7 on Analytical Techniques. Fig. 2.1 shows an IR spectrum.

- (a) Identify the functional group responsible for the broad absorption at 3200–3600 cm⁻¹.

[1]

8 Analytical Techniques — 9701/21/O/N/25/Q8(e) Syllabus 22.1

Question 8 on Analytical Techniques. Fig. 2.1 shows an IR spectrum.

- (a) State the significance of the molecular ion peak in a mass spectrum.

[1]

9 Analytical Techniques — 9701/21/O/N/25/Q9(e) Syllabus 22.1

Question 9 on Analytical Techniques. Fig. 2.1 shows an IR spectrum.

- (a) Identify the functional group responsible for the broad absorption at 3200–3600 cm⁻¹.

[1]

10 Analytical Techniques — 9701/21/O/N/25/Q10(e) Syllabus 22.1

Question 10 on Analytical Techniques. Fig. 2.1 shows an IR spectrum.

- (a) State the significance of the molecular ion peak in a mass spectrum.

[1]

Question Pack

- 11 Analytical Techniques – 9701/21/O/N/25/Q11(e)** Syllabus 22.1
- Question 11 on Analytical Techniques. Fig. 2.1 shows an IR spectrum.
- (a) Identify the functional group responsible for the broad absorption at 3200–3600 cm⁻¹. [1]
- 12 Analytical Techniques – 9701/21/O/N/25/Q12(e)** Syllabus 22.1
- Question 12 on Analytical Techniques. Fig. 2.1 shows an IR spectrum.
- (a) State the significance of the molecular ion peak in a mass spectrum. [1]
- 13 Analytical Techniques – 9701/21/O/N/25/Q13(e)** Syllabus 22.2
- Question 13 on Analytical Techniques. Fig. 2.1 shows an IR spectrum.
- (a) Identify the functional group responsible for the broad absorption at 3200–3600 cm⁻¹. [1]
- 14 Analytical Techniques – 9701/21/O/N/25/Q14(e)** Syllabus 22.2
- Question 14 on Analytical Techniques. Fig. 2.1 shows an IR spectrum.
- (a) State the significance of the molecular ion peak in a mass spectrum. [1]
- 15 Analytical Techniques – 9701/21/O/N/25/Q15(e)** Syllabus 22.2
- Question 15 on Analytical Techniques. Fig. 2.1 shows an IR spectrum.
- (a) Identify the functional group responsible for the broad absorption at 3200–3600 cm⁻¹. [1]
- 16 Analytical Techniques – 9701/21/O/N/25/Q16(e)** Syllabus 22.2
- Question 16 on Analytical Techniques. Fig. 2.1 shows an IR spectrum.
- (a) State the significance of the molecular ion peak in a mass spectrum. [1]
- 17 Analytical Techniques – 9701/21/O/N/25/Q17(e)** Syllabus 22.2
- Question 17 on Analytical Techniques. Fig. 2.1 shows an IR spectrum.
- (a) Identify the functional group responsible for the broad absorption at 3200–3600 cm⁻¹. [1]
- 18 Analytical Techniques – 9701/21/O/N/25/Q18(e)** Syllabus 22.2
- Question 18 on Analytical Techniques. Fig. 2.1 shows an IR spectrum.
- (a) State the significance of the molecular ion peak in a mass spectrum. [1]
- 19 Analytical Techniques – 9701/21/O/N/25/Q19(e)** Syllabus 22.2
- Question 19 on Analytical Techniques. Fig. 2.1 shows an IR spectrum.
- (a) Identify the functional group responsible for the broad absorption at 3200–3600 cm⁻¹. [1]
- 20 Analytical Techniques – 9701/21/O/N/25/Q20(e)** Syllabus 22.2
- Question 20 on Analytical Techniques. Fig. 2.1 shows an IR spectrum.
- (a) State the significance of the molecular ion peak in a mass spectrum. [1]

Question Pack

21 Analytical Techniques – 9701/21/O/N/25/Q21(e) Syllabus 22.2

Question 21 on Analytical Techniques. Fig. 2.1 shows an IR spectrum.

- (a) Identify the functional group responsible for the broad absorption at 3200–3600 cm⁻¹. [1]

22 Analytical Techniques – 9701/21/O/N/25/Q22(e) Syllabus 22.2

Question 22 on Analytical Techniques. Fig. 2.1 shows an IR spectrum.

- (a) State the significance of the molecular ion peak in a mass spectrum. [1]

23 Analytical Techniques – 9701/21/O/N/25/Q23(e) Syllabus 22.2

Question 23 on Analytical Techniques. Fig. 2.1 shows an IR spectrum.

- (a) Identify the functional group responsible for the broad absorption at 3200–3600 cm⁻¹. [1]

24 Analytical Techniques – 9701/21/O/N/25/Q24(e) Syllabus 22.2

Question 24 on Analytical Techniques. Fig. 2.1 shows an IR spectrum.

- (a) State the significance of the molecular ion peak in a mass spectrum. [1]

25 Analytical Techniques – 9701/21/O/N/25/Q25(e) Syllabus 22.2

Question 25 on Analytical Techniques. Fig. 2.1 shows an IR spectrum.

- (a) Identify the functional group responsible for the broad absorption at 3200–3600 cm⁻¹. [1]

Mark Scheme

MARK SCHEME – WEEK 4

Q	MARKING POINTS	MARKS
1(a)	O-H / alcohol (1)	1
2(a)	Gives the relative molecular mass of the compound (1)	1
3(a)	O-H / alcohol (1)	1
4(a)	Gives the relative molecular mass of the compound (1)	1
5(a)	O-H / alcohol (1)	1
6(a)	Gives the relative molecular mass of the compound (1)	1
7(a)	O-H / alcohol (1)	1
8(a)	Gives the relative molecular mass of the compound (1)	1
9(a)	O-H / alcohol (1)	1
10(a)	Gives the relative molecular mass of the compound (1)	1
11(a)	O-H / alcohol (1)	1
12(a)	Gives the relative molecular mass of the compound (1)	1
13(a)	O-H / alcohol (1)	1
14(a)	Gives the relative molecular mass of the compound (1)	1
15(a)	O-H / alcohol (1)	1
16(a)	Gives the relative molecular mass of the compound (1)	1
17(a)	O-H / alcohol (1)	1
18(a)	Gives the relative molecular mass of the compound (1)	1
19(a)	O-H / alcohol (1)	1
20(a)	Gives the relative molecular mass of the compound (1)	1
21(a)	O-H / alcohol (1)	1
22(a)	Gives the relative molecular mass of the compound (1)	1
23(a)	O-H / alcohol (1)	1

Mark Scheme

MARK SCHEME — WEEK 4 (CONTINUED)

Q	MARKING POINTS	MARKS
24(a)	Gives the relative molecular mass of the compound (1)	1
25(a)	O-H / alcohol (1)	1
